

GDSQ-VLA: Geometry- and Distribution-Sensitive Layer Selection for Post-Training Quantization of Vision-Language-Action Models

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Abstract

Post-training quantization can reduce the static storage of vision-language-action (VLA) policies, but uniform precision overlooks the different numerical sensitivity of language reasoning and iterative action generation. We present GDSQ-VLA, a training-free mixed-precision framework that combines geometry and distribution preservation, paired action sensitivity, stability guards, and complete-configuration adjudication to allocate W4/FP16 layers under a frozen byte budget. The final method additionally uses a calibration-gated runtime correction selector: from model-level calibration statistics alone, it chooses the uncorrected policy, attention-temperature modulation (ATM), or output-head balancing (OHB), never their uncalibrated combination. The selector does not use task identity, rollout outcomes, or runtime success feedback. On RoboCasa365, 256-observation equivalence tests are bitwise exact for both evaluated policy families, allowing the GR00T selector’s baseline decision to reuse its frozen 50-task evaluation. This final GR00T configuration obtains 50.8% task-macro success at a theoretical $1.99\times$ candidate-component compression, compared with 55.1% for FP16. The audited $\pi_{0.5}$ selector obtains 27.5% task-macro SR over all 50 tasks and improves over its uncorrected static mask by 2.2 points (95% task CI [0.6, 3.9]). The current evidence supports the mechanism and a favorable GR00T storage-accuracy point, but not yet same-budget superiority across architectures.

1. Introduction

Vision-language-action (VLA) models connect visual perception and language-conditioned reasoning to continuous robot control, allowing one policy to execute diverse instructions across scenes and embodiments [4, 5, 10, 14, 25]. Diffusion- and flow-based action heads improve continuous action generation [3, 7, 20, 24, 26], but large multimodal backbones and iterative denoising make deployment storage and memory bandwidth important constraints. Post-training

quantization (PTQ) is appealing because it changes numerical precision without retraining the policy or redesigning its operator graph [18, 35].

VLA quantization is difficult for two related reasons. First, numerical sensitivity is heterogeneous: an error in a language layer changes multimodal conditioning, whereas an error in an action layer can accumulate across denoising steps. Uniform low precision ignores this structure, while a fixed architecture-level protection rule cannot adapt to a new checkpoint. Second, a mixed-precision mask and its runtime correction are coupled. Attention-temperature modulation (ATM) and output-head balancing (OHB) may compensate residual quantization drift, but applying either mechanism indiscriminately can perturb an already stable model; applying both can double-correct the same propagated error. A deployable method must therefore decide both *where* to quantize and *whether* a calibrated correction is warranted.

Existing mixed-precision methods rank layers using local reconstruction error, activation magnitude, Hessian sensitivity, or task-specific proxies [6, 8, 9, 12, 13, 41]. These signals are useful but incomplete for generative control. Centered kernel alignment (CKA) measures preservation of inter-sample geometry [15, 22, 27], yet is intentionally insensitive to isotropic scaling. Cauchy-Schwarz (CS) divergence measures distribution overlap [38], but does not say whether a shift changes the executed action. Neither local metric captures cancellation or amplification when many layers are quantized together, and neither determines whether a runtime correction should be admitted.

We address these challenges with GDSQ-VLA, a training-free W4/FP16 allocation and correction-selection framework. For each candidate Linear layer, an isolated W4A8 intervention is compared with a paired full-precision attribution path. CKA and CS provide complementary representation evidence; paired action-trajectory sensitivity weights that evidence; and activation-RMS and saturation guards reject infeasible assignments. A byte-constrained allocator generates diverse masks, after which complete-configuration functional adjudication measures cross-layer

action error. The result is a frozen mixed-precision plan rather than a per-task policy.

The final method then applies a *calibration-gated runtime correction selector*. From model-level statistics of the frozen correction artifact, three preregistered thresholds either reject correction or admit the single dominant mechanism. Its output is restricted to the uncorrected baseline, ATM, or OHB; ATM+OHB is never an allowed final decision. The selector cannot inspect task identity, rollout success, or runtime feedback. The frozen v8 rule selects the baseline for GR00T N1.5 and OHB for $\pi_{0.5}$. On 256 frozen observations per model, each runtime path is bitwise identical to its corresponding static path, providing an explicit gate for any result reuse.

We evaluate on RoboCasa365 [23] with official horizons, four denoising steps, 16 executed actions per replan, and paired action noise. The completed GR00T selector result obtains 50.8% task-macro success over 50 tasks at a theoretical $1.99\times$ candidate-component compression, versus 55.1% for FP16. The original QuantVLA configuration obtains 30.4%, but uses a different compression rate and is therefore a secondary comparison rather than same-budget causal evidence. The final $\pi_{0.5}$ selector obtains 27.5% all-task macro SR. It improves over the uncorrected static mask (25.3%) by 2.2 points (95% task CI [0.6, 3.9], Holm-adjusted $p = 0.045$). Its 1.3-point difference from FP16 has CI [-0.5, 3.2] and Holm-adjusted $p = 0.240$, so this comparison is competitive rather than superior. Both uniform-W6 controls have complete audited coverage and are reported descriptively. The search-matched random and action-only controls remain excluded from superiority claims until their exact manifests reach full coverage.

Our contributions are threefold:

- (1) We formulate VLA precision allocation as action-weighted preservation of representation geometry and distribution overlap, with explicit stability guards and complete-mask functional adjudication under a frozen byte budget.
- (2) We introduce a model-level calibration gate that selects at most one runtime correction mechanism without task metadata or outcome feedback, and we verify its selected execution paths by bitwise equivalence.
- (3) We provide an auditable RoboCasa365 protocol in which manifests bind checkpoints, plans, code, environments, task/seed coverage, and paper cells; unsupported or incomplete results cannot populate the main table.

2. Related Work

Efficient VLA inference. Generalist VLA policies combine large multimodal encoders with continuous-control heads [3, 4, 14, 24–26]. Efficiency methods reduce their cost through compact architectures, dynamic depth, token reduction, or temporal reuse [30, 36, 37, 40]. SQAP-VLA co-

designs quantization-aware token pruning with the quantizer to obtain structured inference acceleration [11]. These directions can reduce computation as well as storage, but modify architecture, token flow, or execution. Our allocation leaves the checkpoint and operator graph intact; consequently, our present claim is static low-bit storage rather than latency acceleration.

VLA post-training quantization. QuantVLA adapts DuQuant-style transformations to language and diffusion-action stacks and calibrates attention temperature and output-head energy [39]. ActQuant is the closest action-guided mixed-precision method: it assigns tensor bit widths from action contribution and optimizes block scales with action-aware curvature, targeting sub-4-bit deployment [1]. Ω -QVLA instead pursues full-stack W4A4 quantization of both the language backbone and diffusion head using composite rotations and denoising-step scaling [33]. These methods establish that control signals and the action head require special treatment. GDSQ-VLA addresses a different question: which layers should retain FP16 under a fixed byte budget when local representation damage, downstream action sensitivity, and complete-mask interactions disagree. We preregister a pinned-code LIBERO reproduction and report only our locally audited values before making any shared-protocol ranking.

Transformer and diffusion PTQ. Weight-only methods such as GPTQ and AWQ optimize rounding or protect activation-salient channels [12, 19]. Weight-activation PTQ reshapes difficult distributions through smoothing, calibration, or rotations [2, 18, 21, 29, 31, 35]. Diffusion PTQ additionally handles timestep-dependent activation ranges and iterative error accumulation [6, 16, 28, 34, 41]. These techniques improve the low-bit numerical substrate. We reuse a DuQuant-style W4A8 substrate so that the studied variable is precision allocation and correction selection, not a new quantizer kernel.

Mixed precision and representation evidence. Mixed-precision allocation has been driven by hardware feedback, Hessian sensitivity, reconstruction loss, and task-specific diffusion metrics [8, 9, 13, 17, 32, 41]. SVCCA and projection-weighted CCA compare representation subspaces [22, 27], while CKA measures sample geometry with invariance to orthogonal transformations and isotropic scaling [15]. CS divergence provides a complementary distribution-overlap statistic [38]. GDSQ-VLA uses these metrics as frozen intervention probes, scales them with paired action sensitivity, and reserves action-functional evaluation for complete masks. Its runtime selector is separate from mask search: a preregistered model-level gate admits at most one calibrated correction and cannot use task identity or rollout outcomes.

3. Method

Overview. GDSQ-VLA converts a pretrained VLA policy into a frozen mixed-precision policy and a model-level correction decision without updating model parameters. The allocation pipeline in Fig. 1 has four stages: isolated layer interventions, dual-similarity and action scoring, constrained mask search, and paired functional adjudication. A fifth stage calibrates ATM and OHB for the selected mask and applies a preregistered absolute gate. The deployed output is therefore a pair (M, v) , where M is a W4/FP16 mask and $v \in \{\text{baseline}, \text{ATM}, \text{OHB}\}$ is fixed for the model.

3.1. Isolated Quantization Interventions

Given visual observations I , instruction l , robot state s , and denoising time t , a VLA policy iteratively updates an action sample

$$x_{t-1} = f_{\theta}(x_t, F_{VL}(I, l), s, t). \quad (1)$$

We use group-wise DuQuant-style W4 weights and static A8 activations with group size 64 [18]. The selectable inventory is checkpoint-specific. GR00T contains 116 candidate Linear layers—84 language projections and 32 DiT feed-forward projections—while 64 protected DiT attention projections remain FP16. The $\pi_{0.5}$ inventory contains 180 language and action-expert Linear layers.

A single reference path cannot separate W4 attribution from wrapper mechanics. We therefore use an *attribution reference* R that retains the transformations and A8 wrappers but routes every candidate weight through its full-precision branch. The *deployment reference* is the unwrapped FP16 policy. For candidate i , all layers follow R except i , which uses W4. A fixed calibration buffer, token mask, and initial action noise are shared between the intervention and R . Hooks collect matched outputs (H_i^R, H_i^Q) ; padding rows are removed with the reference mask, and deterministic stratified pooling prevents visual, language, or state tokens from dominating. Representation and single-layer action scores use R , whereas activation guards and complete-mask adjudication use the deployment reference.

3.2. Dual Similarity, Action Weighting, and Guards

CKA measures preservation of inter-sample geometry. For centered matrices $\tilde{H}_i^R, \tilde{H}_i^Q \in \mathbb{R}^{N \times d}$, let $G_i^{UV} = (\tilde{H}_i^U)^{\top} \tilde{H}_i^V$. We compute

$$\text{CKA}_i = \frac{\|G_i^{RQ}\|_F^2}{\|G_i^{RR}\|_F \|G_i^{QQ}\|_F + \epsilon}, \quad D_i^{\text{CKA}} = 1 - \text{CKA}_i. \quad (2)$$

Its invariance to isotropic scaling is useful for stable geometry comparison but creates a scale blind spot. We complement it with an empirical CS divergence. For Gaussian kernel k whose bandwidth is fixed from the reference sam-

ples,

$$D_i^{\text{CS}} = \log \left(\frac{1}{N^2} \sum_{a,b} k(h_a^R, h_b^R) \right) + \log \left(\frac{1}{N^2} \sum_{a,b} k(h_a^Q, h_b^Q) \right) - 2 \log \left(\frac{1}{N^2} \sum_{a,b} k(h_a^R, h_b^Q) \right). \quad (3)$$

We use log-sum-exp accumulation and a bandwidth floor of 10^{-3} . After joint min-max normalization across candidates, the representation score is

$$S_i = \lambda_{\text{CKA}} \overline{D^{\text{CKA}}}_i + \lambda_{\text{CS}} \overline{D^{\text{CS}}}_i. \quad (4)$$

Representation damage matters only when it changes control. Under the same isolated intervention and paired denoising noise, let d_i denote action-trajectory divergence from R . We define

$$w_i = \frac{nd_i}{\sum_j d_j + \epsilon}, \quad (5)$$

then robustly clip and renormalize the weights so that $n^{-1} \sum_i w_i = 1$. This distinguishes layers with similar representation distortion but different downstream effects.

Similarity is not a stability certificate. Each intervention also records median log-RMS drift from unwrapped FP16 and the estimated fraction beyond the downstream static-A8 range. GR00T uses the frozen thresholds $\tau_{\text{RMS}} = 0.1319$ and $\tau_{\text{sat}} = 0.00232$; $\pi_{0.5}$ uses its separately calibrated thresholds. A violation forces FP16. The saturation statistic is an engineering approximation based on the calibrated consumer scale, not a direct measurement of fused-kernel clipping.

3.3. Byte-Constrained Search and Functional Adjudication

Let $x_i = 1$ assign W4A8 and $x_i = 0$ preserve the original FP16 Linear. Candidate masks minimize

$$\min_x \sum_i x_i w_i S_i, \quad (6)$$

$$\text{s.t.} \quad \sum_i [x_i C_i^{\text{W4}} + (1 - x_i) C_i^{\text{FP16}}] \leq B,$$

$$x_i = 0 \quad \text{for every guard violation,} \quad x_i \in \{0, 1\}.$$

Here C_i is theoretical tightly packed weight storage. The uniform-W6 candidate-scope budgets are 862,912,512 bytes for GR00T and 2,042,542,080 bytes for $\pi_{0.5}$. They exclude activations, CUDA workspaces, simulator state, and fake-quant caches. Greedy allocation and local bit flips produce diverse low-objective masks; the additive optimum is not treated as the final policy.

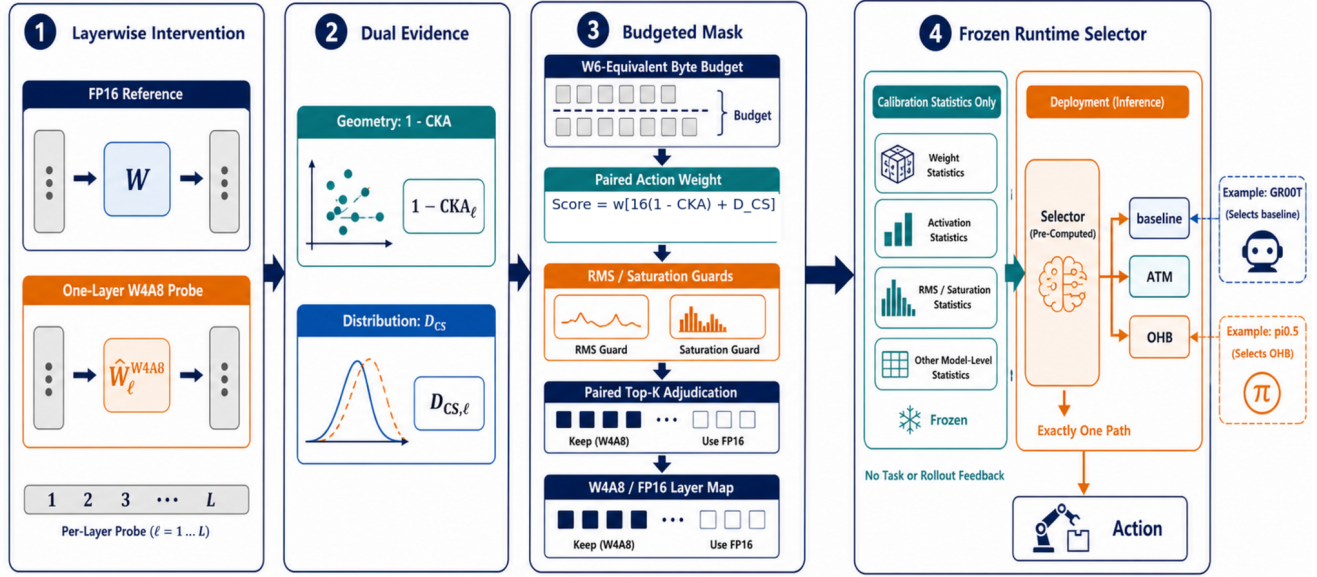


Figure 1. GDSQ-VLA from frozen calibration to action generation. Single-layer interventions measure complementary geometric and distributional changes; action weighting, feasibility guards, a W6-equivalent byte budget, and paired functional adjudication freeze the W4/FP16 mask. A separate model-level calibration gate then selects exactly one of baseline, ATM, or OHB without task identity or rollout feedback.

Simultaneous interventions can cancel or amplify. We therefore instantiate selected Top- K masks with plan-specific A8 calibration and compare complete policies under paired observations and noise. The functional score is

$$D_{\text{func}} = D_{\text{final}} + D_{\text{kin}} + D_{\text{grip}} + 2 \text{CVaR}_{0.9}(D_{\text{solver}}), \quad (7)$$

where the tail term prevents a low mean from hiding severe denoising failures. Functional adjudication selects a representative for each CKA:CS ratio. On four predeclared GR00T development tasks, the frozen rule selects 16:1 from $\{8, 16, 32, 64, 128, 256\} : 1$ and a CKA-only boundary. The final masks assign W4 to 100 of 116 GR00T candidates and 80 of 180 $\pi_{0.5}$ candidates. A 10,000-draw task/seed bootstrap and leave-one-task-out analysis conducted after freezing finds no ratio whose paired interval against 16:1 excludes zero; we therefore interpret 16:1 as a selected point on a broad development plateau, not a universal optimum.

3.4. Calibration-Gated Runtime Correction

The mixed-precision mask can leave structured attention drift, but a correction can also destabilize an already adequate policy. For the frozen mask, ATM calibrates a query-temperature multiplier α and OHB calibrates per-head output multipliers β_h . We first choose one *candidate* mechanism from their model-level calibration geometry. OHB is the candidate when its correction is broad and its mean absolute log-magnitude is at least twice that of ATM. ATM is the candidate when corrected attention projections remain FP16

and its directional coherence exceeds OHB by at least 0.05. Otherwise the candidate is the baseline. ATM and OHB are not composed because their ratios act at different points of the same propagated attention error and are not calibrated as an additive correction.

The v8 absolute gate then decides whether to admit the candidate. For correction values $c_1, \dots, c_m$, define

$$q_{\text{dir}} = \frac{|n_{<1} - n_{>1}|}{m}, \quad q_{\text{non}} = \frac{n_{c \neq 1}}{m}, \quad q_{\text{log}} = \frac{1}{m} \sum_j |\log c_j|. \quad (8)$$

The correction is enabled only if $q_{\text{dir}} \geq 0.30$, $q_{\text{non}} \geq 0.85$, and $q_{\text{log}} \geq 0.05$; otherwise the baseline is selected. These three thresholds and the selector SHA were frozen before formal testing. GR00T’s ATM candidate fails all three gates and therefore selects the uncorrected GDSQ mask. The $\pi_{0.5}$ OHB candidate passes all three and selects OHB. The decision is evaluated once per model, so task names retained in runtime metadata are attestations only and cannot affect the selected variant.

Every formal result row records the selector SHA, rule name, selected variant, selected configuration ID, and server-metadata hash. For GR00T, whose bitwise gate permits reuse of the immutable static matrix, a separate 2,500-row derived attestation adds these selector fields and the source-row hash without modifying the original outcomes. The runtime rejects an unknown variant and v8 rejects ATM+OHB.

Consequently, the final method is exactly GDSQ-VLA plus one frozen model-level choice from {baseline, ATM, OHB}; static GDSQ, ATM, OHB, and ATM+OHB configurations are ablations.

4. Experiments

4.1. Experimental Setup

Benchmark and policies. We evaluate GR00T N1.5 and $\pi_{0.5}$ [26] on the RoboCasa365 v1.0.1 target split [23]. The benchmark has 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks. GR00T uses the corresponding 60k-step post-training checkpoints; $\pi_{0.5}$ uses its RoboCasa checkpoint. Four atomic tasks—OpenCabinet, OpenStandMixerHead, PickPlaceDrawerToCounter, and CoffeeSetupMug—form the predeclared GR00T ratio-development set. The $\pi_{0.5}$ allocation transfers the frozen 16:1 ratio and performs model-specific functional adjudication and calibration; it does not tune the ratio from $\pi_{0.5}$ test rollouts.

Configurations and budgets. The final configuration is GDSQ-VLA plus the frozen v8 runtime selector. The selector chooses the uncorrected GDSQ mask for GR00T and OHB for $\pi_{0.5}$. Static GDSQ, ATM, OHB, and ATM+OHB are ablations, not alternative definitions of ours. FP16 is the uncompressed reference. QuantVLA W4A8 [39] is retained as a secondary low-bit comparison because its compression rate differs from ours. The direct budget controls are uniform W6, a search-matched random W4/FP16 mask, and an action-only allocator. Random search uses the same 26 candidates, functional-evaluation count, and seven development representatives as GDSQ; its seeds and masks are frozen before held-out rollouts. All masks obey the same uniform-W6 candidate-byte ceiling and use the same checkpoint, A8 calibration strategy, and guards.

Rollout and statistics. Every formal configuration uses 50 seeds per task, official horizons, a fresh rendered environment, four denoising steps, and 16 executed actions per replan. Initial action noise is deterministically keyed by (task, seed, replan), giving paired policy comparisons without changing the marginal normal distribution. We report task-macro SR, 10,000-draw task-cluster bootstrap intervals, paired task-level sign-flip tests, and Holm correction over preregistered contrasts. A no-guards crash, NaN, or out-of-range action is a formal failure rather than a silently dropped episode.

Claim–evidence gate. Each run manifest binds checkpoint, plan, calibration, selector, launcher, evaluator, environment, task list, and seed coverage by SHA-256. Each

result row records its server-metadata hash and, when applicable, selector SHA, rule, variant, and configuration ID. The main table is generated from the evidence registry; a task-set cell remains pending unless its declared coverage is exact, duplicates are absent, statistics are reproducible, and source drift has an explicit audit. Replay audits retain the frozen launch manifests: they reconstruct every task/seed key and must reproduce the registered summary exactly before later source edits can be disclosed without invalidating immutable result rows. This gate is passed by the completed $\pi_{0.5}$ four-configuration matrix, the three-manifest GR00T uniform-W6 evaluation, and the independently complete GR00T Ω -QVLA Atomic subset.

External Table 1 reproduction. For Ω -QVLA [33], we pin the authors’ code commit but rebuild task-set-specific W4A4 packs from frozen RoboCasa365 calibration data because the released LIBERO packs cannot be transferred. Table 1 exposes independently gated task-set cells: the GR00T Atomic subset is complete, whereas its Composite cells and all $\pi_{0.5}$ cells remain pending. Table 2 separately uses the same five configurations as Table 1—FP16, QuantVLA W4A8, Uniform W6, Ω -QVLA W4A4, and ours—under the original LIBERO protocol: four suites, ten tasks per suite, ten held-out trials per task, and eight denoising steps. We import no author-reported values; all 4,000 local episodes must pass the joint manifest gate.

4.2. Selector Validation

The path-equivalence test uses 256 frozen observations per model, four denoising steps, and paired noise. GR00T compares the selector’s baseline decision with static GDSQ; $\pi_{0.5}$ compares the selector’s OHB decision with static OHB. Both comparisons are bitwise equal on all 256/256 action tensors, with maximum absolute difference zero and no non-finite output. This passes the preregistered reuse gate for the completed GR00T static 50-task run. It does *not* authorize reuse of a partial $\pi_{0.5}$ OHB diagnostic, so we run a separate formal 50-task selector evaluation for $\pi_{0.5}$.

Before formal evaluation, a two-task, two-seed closed-loop preflight verifies four complete episodes and all row-level attestations; this preflight is diagnostic only. The $\pi_{0.5}$ manifest is then frozen before the first official rollout and specifies exactly 2,500 unique task/seed keys. The audited run completes all 2,500 manifested episodes; its three task-group macro SRs are 57.8%, 17.0%, 3.9%, for 27.5% over all 50 tasks. All 2,500 rows select OHB and attest the frozen selector SHA; no task metadata or runtime outcome enters the decision.

4.3. Completed RoboCasa365 Evidence

GR00T final selector. By bitwise reuse, the final GR00T selector obtains 69.0%, 40.6%, and 40.4% task-macro SR on

Table 1. RoboCasa365 task-macro success rate (SR, %) over 50 paired scenarios per task.

Configuration	Low-bit layers	Atomic SR ↑	C-Seen SR ↑	C-Unseen SR ↑	All SR ↑	Size (GiB) ↓	Comp. ↑
GR00T N1.5 (<i>target split</i>)							
FP16	—	75.6	41.9	45.3	55.1	1.993	1.00×
QUANTVLA W4A8	116 W4	49.6	19.6	19.8	30.4	0.898	2.22×
Uniform W6	116 W6	68.4	42.9	41.8	51.7	1.109	1.80×
Ω -QVLA W4A4 [‡]	180 W4	60.1	<i>pending</i>	<i>pending</i>	<i>pending</i>	0.599	3.33×
GDSQ-VLA(Ours)	100 W4	69.0	40.6	40.4	50.8	1.001	1.99×
$\pi_{0.5}$ [26] (<i>target split</i>)							
FP16	—	58.3	14.0	2.1	26.2	4.113	1.00×
QUANTVLA W4A8	180 W4	56.1	12.8	1.8	24.8	1.388	2.96×
Uniform W6	180 W6	56.6	10.9	2.1	24.5	1.902	2.16×
Ω -QVLA W4A4 [‡]	W4A4	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
GDSQ-VLA(Ours)	80 W4	57.8	17.0	3.9	27.5	1.868	2.20×

C-Seen/C-Unseen denote Composite-Seen/Composite-Unseen. Ours uses the frozen v8 selector. Uniform W6 is the direct byte-budget baseline; QuantVLA is a different-rate comparison. Size is theoretical tightly packed candidate-Linear storage, with FP16 shown as the reference. Pending cells are claim-disabled. [‡] Ω -QVLA uses RoboCasa365-specific calibration; its GR00T Atomic cell is complete, while the remaining cells await exact coverage. Its displayed size includes packed W4 weights and required scales/rotation metadata, not the dequantized evaluation pack.

Atomic, Composite-Seen, and Composite-Unseen, respectively, for 50.8% over all 50 tasks. FP16 obtains 55.1%; the paired all-task difference for ours minus FP16 is -4.32 points with a 95% task-cluster interval of $[-6.56, -2.08]$. The QuantVLA configuration obtains 30.4%, and ours exceeds it by 20.3 points with interval $[16.9, 24.0]$. Because QuantVLA compresses the studied scope by $2.22\times$ while ours compresses it by $1.99\times$, this is a descriptive secondary comparison, not evidence that the selection signal is superior at fixed storage.

The completed uniform-W6 control obtains 68.4%, 42.9%, and 41.8% on the three task groups, for 51.7% overall at $1.80\times$ theoretical compression. Ours is -1.0 point relative to W6, with a paired 95% task-cluster interval of $[-3.1, 1.2]$, while using 1.001 rather than 1.109 GiB in the candidate Linear scope. We therefore treat the GR00T fixed-ceiling comparison as competitive storage-accuracy evidence, not superiority.

Ω -QVLA partial RoboCasa365 evidence. The completed GR00T Atomic evaluation contains exactly 900 episodes—18 tasks with 50 paired seeds each—and no formal failure. Ω -QVLA obtains 60.1% task-macro SR, with a 95% task-cluster interval of $[50.7, 68.7]$. Its theoretical tightly packed Linear-component size is 0.599 GiB, or $3.33\times$ relative to the same 1.993-GiB FP16 scope. Composite-Seen, Composite-Unseen, and the 50-task mean remain pending, so this subset does not support an all-task ranking.

$\pi_{0.5}$ final selector and architecture boundary. The final $\pi_{0.5}$ selector obtains 27.5% all-task macro SR. It improves

over the uncorrected static mask (25.3%) by 2.2 points (95% task CI $[0.6, 3.9]$, Holm-adjusted $p = 0.045$). Its 1.3-point difference from FP16 has CI $[-0.5, 3.2]$ and Holm-adjusted $p = 0.240$, so this comparison is competitive rather than superior. The selector also exceeds QuantVLA W4A8 by 2.6 points (95% task CI $[0.9, 4.5]$, Holm-adjusted $p = 0.023$), but QuantVLA has a different $2.96\times$ compression rate and remains a secondary comparison. The three task-group SRs of the final selector are 57.8%, 17.0%, and 3.9%, yielding 27.5% over all 50 tasks at $2.20\times$ theoretical candidate-component compression.

The uncorrected static mask obtains 25.3%, while static ATM+OHB reaches 26.6% as an ablation. The selector’s significant gain over the same frozen mask isolates the value of admitting OHB for this model; it does not establish that the allocation is universally superior. The completed uniform-W6 control obtains 24.5% at $2.16\times$ theoretical compression, giving a descriptive ours-minus-W6 gap of 3.0 points at nearly matched storage. We do not interpret this gap as superiority before the remaining same-budget controls and preregistered joint Holm correction are complete. In particular, the final selector’s 1.3-point difference from FP16 has an interval crossing zero. We therefore report $\pi_{0.5}$ as an architecture boundary: runtime correction helps the transferred allocation, while fixed-budget superiority remains unproven.

Same-budget causal controls. Uniform W6, shown in both main tables, isolates whether mixed precision helps under the same byte ceiling. Random masks control candidate count, functional-search budget, and retained FP16 bytes. The action-only allocator tests whether paired behavioral sensitivity alone explains the selected layers. The result-

blind functional screens for both search controls are complete, but their closed-loop development selection and formal held-out rollouts are not. These two search-mechanism controls remain supplemental because they answer causal questions rather than define deployable headline configurations; they will be reported only after complete audited manifests. Superiority wording additionally requires the paired 95% interval for ours minus the strongest same-budget control to exclude zero under the preregistered Holm correction. Both uniform-W6 controls have complete audited coverage and are reported descriptively. The search-matched random and action-only controls remain excluded from superiority claims until their exact manifests reach full coverage.

4.4. LIBERO Comparison

The reproduction uses the pinned release and official quantization packs for Ω -QVLA and independently calibrates QuantVLA, Uniform W6, and GDSQ-VLA on the original GR00T N1.5 and $\pi_{0.5}$ checkpoints. We freeze the commands before observing local rollouts, use initialization offset 10 for the ten held-out states, and recalibrate the GDSQ selector per model without reusing RoboCasa artifacts. The audit preserves two release discrepancies: the paper and current builder documentation use different GPTQ damping values, and the released $\pi_{0.5}$ pack has ten per-step buckets while the evaluation path uses eight denoising steps. We load the released packs as-is without rebuilding or test-set retuning.

4.5. Ratio Stability and Component Tests

The frozen point rule selects CKA:CS= 16:1 at 80.5% development SR. Post-freeze hierarchical task/seed bootstrap gives it the largest fractional best probability (0.486), but every alternative’s paired interval against 16:1 includes zero. Leave-one-task-out selects 16:1 on three omissions and ties it with 256:1 on the fourth. We therefore describe a broad ratio plateau rather than sharp 16:1 optimality. The ratio table, selection-controlled Primary-14 comparison, and earlier five-seed signal diagnostic are moved to the supplement. The new 50-seed component matrix—CKA-only, CS-only, $w_i = 1$, no guards, and no functional adjudication—is preregistered on GR00T Primary-14; incomplete cells cannot replace those older diagnostics.

4.6. Storage and Runtime Scope

The GR00T final plan stores an estimated 1.001 GiB over the LLM+DiT Linear scope, versus 1.993 GiB for FP16, giving $1.99\times$ theoretical compression. The $\pi_{0.5}$ plan stores 1.868 GiB versus 4.113 GiB. The GR00T Ω -QVLA estimate is 0.599 GiB and additionally includes its required DiT permutation indices and per-step A4 scale tables. These estimates include tightly packed low-bit weights, weight scales, block rotations, and native FP16 for protected layers and biases. They exclude the vision stack, activations, CUDA

workspaces, and simulator state. The current Ω -QVLA evaluation artifact is a 6.41-GiB dequantized FP16 pack with dense rotations; it validates the W4A4 numerical path but is not the packed-deployment size reported in Table 1.

The current execution path is eager fake quantization: it retains floating-point weights and executes floating-point GEMMs. On GR00T it therefore takes 0.254 seconds per replan and peaks at 17.7 GiB, compared with 0.139 seconds and 12.8 GiB for FP16. We make no latency or live-memory compression claim. Packed checkpoint size, isolated batch-1 policy-server latency, and process memory will be reported separately; end-to-end deployment gains require fused INT4/INT8 kernels.

5. Conclusion and Limitations

This paper presents GDSQ-VLA, a training-free framework that couples action-weighted geometry/distribution preservation with byte-constrained W4/FP16 allocation and a calibration-gated runtime correction selector. The allocation uses CKA, CS divergence, stability guards, and complete-mask functional adjudication; the selector makes one frozen model-level choice among baseline, ATM, and OHB without task identity or outcome feedback. Bitwise path tests pass on 256 frozen observations for both policy families. The completed GR00T selector obtains 50.8% task-macro SR over 50 RoboCasa365 tasks at a theoretical $1.99\times$ candidate-component compression, compared with 55.1% for FP16. The final $\pi_{0.5}$ selector obtains 27.5% all-task macro SR. It improves over the uncorrected static mask (25.3%) by 2.2 points (95% task CI [0.6, 3.9], Holm-adjusted $p = 0.045$). Its 1.3-point difference from FP16 has CI [-0.5, 3.2] and Holm-adjusted $p = 0.240$, so this comparison is competitive rather than superior.

The evidence also defines clear boundaries. The large GR00T and $\pi_{0.5}$ differences from QuantVLA are secondary, different-compression comparisons. Same-budget controls must complete before supporting a causal superiority claim. The 16:1 ratio lies on a broad four-task development plateau and is transferred, rather than independently selected by $\pi_{0.5}$ rollouts. Finally, the present fake-quant path demonstrates theoretical static storage only: it does not reduce live GPU memory or latency.

Future work will recalibrate and evaluate the method on all four LIBERO suites, measure mask and selector stability across calibration sizes and prompts, trace a frozen-budget storage—SR Pareto frontier, and report packed checkpoint size, isolated policy-server memory, and batch-1 latency. Shared-protocol comparisons with ActQuant, HoloQ-VLA, and SQAP-VLA require reproducible official implementations; otherwise, the preregistered action-only control remains the appropriate internal baseline. Fused INT4/INT8 kernels and real-robot evaluation are necessary before making an end-to-end deployment claim.

Table 2. LIBERO local comparison under the Ω -QVLA Table 1 protocol (success rate, %).

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$
GR00T N1.5					
FP16	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
QUANTVLA W4A8	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Uniform W6	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Ω -QVLA W4A4	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
GDSQ-VLA(Ours)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
$\pi_{0.5}$					
FP16	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
QUANTVLA W4A8	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Uniform W6	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Ω -QVLA W4A4	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
GDSQ-VLA(Ours)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

Four official suites, ten tasks per suite, and ten held-out trials per task. All cells remain pending until the joint five-configuration manifest reaches exactly 4,000 valid local episodes; no author-reported values are imported. The official Ω -QVLA W4A4 packs are pinned. QuantVLA, Uniform W6, and GDSQ-VLA are independently calibrated on LIBERO; the GDSQ selector is model-specific, with no RoboCasa artifact reuse or test-set retuning.

Table 3. Observed eager fake-quant runtime. These measurements explain why we claim theoretical component storage rather than end-to-end speed or live-memory gains. Green cells compare compressed configurations only; FP16 is an uncompressed reference.

Configuration	Wall (s)	Replan (s)	Peak MiB
FP16 (uncompressed)	26.9	0.139	12,769
QuantVLA W4A8	37.5	0.268	17,973
GDSQ-VLA +ATM/OHB	31.6	0.259	17,719
GDSQ-VLA (Ours)	30.9	0.254	17,671

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Table 4. Predeclared ratio selection on four development tasks (50 scenarios per task). Lower D_{func} is better, but the final ratio is selected by closed-loop macro success rate.

CKA:CS ratio	$D_{\text{func}} \downarrow$	Dev SR (%) $\uparrow$
8:1	0.11597	77.0
16:1	0.10044	80.5
32:1	0.13395	76.5
64:1	0.07495	75.0
128:1	0.10236	77.0
256:1	0.09328	79.5
CKA only	0.09157	77.5

Table 5. Selection-controlled GR00T N1.5 Atomic results. The CKA:CS ratio is frozen using four development tasks; none of the Primary 14 tasks contributes to ratio selection. The final selector chooses the baseline path and is bitwise identical to the static GDSQ-VLA row. Confidence intervals are 95% task-cluster bootstrap intervals.

Configuration	Primary 14 SR (%) $\uparrow$
FP16 (uncompressed)	73.9 [66.0, 80.9]
QUANTVLA W4A8	45.6 [34.4, 57.0]
GDSQ-VLA + ATM/OHB	63.0 [52.0, 72.9]
GDSQ-VLA + selector	65.9 [57.3, 73.7]

A. Supplementary Evidence

A.1. Ratio Sweep and Stability

The point-selection rule freezes 16:1 because it has the largest development task-macro SR. A separate post-freeze analysis resamples tasks and then paired seeds for 10,000 draws. No alternative ratio has a paired difference interval against 16:1 that excludes zero: for example, the 256:1 interval is $[-6.5, 5.0]$ points and the CKA-only interval is $[-10.0, 3.5]$ points. Leave-one-task-out chooses 16:1 for three omissions and ties 16:1 with 256:1 for the fourth. These diagnostics support stability within a plateau, not universal ratio optimality.

A.2. Selection-Controlled GR00T Atomic Result

The four ratio-development tasks are excluded from this comparison. The final GR00T selector chooses the baseline path and is bitwise identical to the static GDSQ-VLA row. It obtains 65.9% SR on Primary-14, versus 45.6% for QuantVLA W4A8; the paired difference is +20.3 points with interval $[13.9, 26.9]$. FP16 obtains 73.9%. Because QuantVLA and GDSQ-VLA use different static byte totals, the result is a selection-controlled task comparison but not a fixed-budget causal comparison.

Table 6. Preregistered GR00T component ablations on Primary-14 with 50 paired seeds per task. Δ is full GDSQ-VLA minus the ablation; p is Holm-adjusted over five contrasts. Crashes, NaNs, and out-of-range actions in the no-guards arm count as failures.

Config.	SR (%) Δ [95% CI]	Holm p	Fail. (%)
Full GDSQ-VLA	65.9	—	—
CKA only	<i>pending</i>	<i>pending</i>	<i>pending</i>
CS only	<i>pending</i>	<i>pending</i>	<i>pending</i>
Uniform weights ($w_i = 1$)	<i>pending</i>	<i>pending</i>	<i>pending</i>
No guards	<i>pending</i>	<i>pending</i>	<i>pending</i>
No adjudication	<i>pending</i>	<i>pending</i>	<i>pending</i>

Table 7. Five-seed diagnostic ablation. This screen predates the 50-scenario ratio freeze and therefore studies signal composition rather than the final 16:1 mask. Green cells compare quantized configurations only.

Configuration	Held-out 14 SR	All 18 SR
FP16 (uncompressed)	71.4	73.3
W4A8 + ATM/OHB	52.9	52.2
CS only	58.6	61.1
CKA only	57.1	60.0
CKA + ATM/OHB	60.0	63.3
CS+CKA (64:1)	67.1	67.8

A.3. Preregistered 50-Seed Component Ablations

The table is populated only after all five configurations reach exact Primary-14 $\times$ 50 coverage and the five full-versus-ablation contrasts receive a joint Holm correction. Until then, its pending cells are claim-disabled. In the no-guards arm, a NaN, crash, or out-of-range action is retained as a formal failure rather than silently dropped.

A.4. Earlier Five-Seed Diagnostic

This screen predates the 50-seed ratio freeze. Its overlapping task-cluster intervals permit only a qualitative observation that CKA and CS can be complementary. It does not substitute for the preregistered Primary-14 $\times$ 50 component ablation containing CKA-only, CS-only, $w_i = 1$, no guards, and no functional adjudication.

A.5. Static Correction Ablation

On the complete GR00T 50-task matrix, static ATM+OHB changes task-macro SR from 50.8% to 49.4%; the paired interval $[-3.28, 0.52]$ points includes zero. On the complete $\pi_{0.5}$ static matrix, ATM+OHB changes 25.3% to 26.6%, while the preregistered held-out interval $[-0.39, 3.13]$ and Holm-adjusted $p = 0.542$ do not establish a gain. Neither combined configuration is an allowed output of the v8 selector. Standalone ATM and OHB remain static ablations; only the selector-chosen OHB runtime path populates the

817 final row, and it does so through a separate complete 2,500-
818 episode manifest.